

Multi-Modal Migraine Relief Headset System

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Abstract—Migraine is a very common neurological disorder which is usually aggravated by stress, muscle tension, sensory overload and abnormal breathing. Current drug-based interventions offer relief although they might have side effects or they may not treat underlying triggers. In order to provide a safe, non-invasive, and customizable solution, this paper provides the design and development of a Multi-Modal Migraine Relief Headset System, an IoT-enabled therapeutic wearable to combine sound therapy, vibration/massage therapy, and guided breathing exercises into a single hardware-software solution. The technology comprises a headphone-based sound module to provide relaxing soundscapes, a headband fitted on the forehead with coin vibration motors to target the muscle relaxation, and a web-based application that receives the migraine parameters reported by the user (e.g. the intensity of the pain, stress levels, the location of the pain, and the symptomatology of the pain) and creates a tailored therapy session. Arduino-powered actuator unit allows the vibration to be adjusted in real-time by using PWM whereas the web interface leads the user on a sequence of 3-4-3 breathing pattern and subsequent sound therapy optimization. Users are further able to define the vibration modes and the intensity manually according to the comfort and therapeutic values. The solution suggested is multi-modal drug free and adaptive in the reduction of migraine. The system will focus on lowering the severity of symptoms, enhancing comfort, and self-treatment of migraine factors, making it possible to help a patient long-term by combining physiological relaxation methods with individualized electronic control. This paper proves that it is possible to incorporate wearable technology and smart software to make a natural migraine solution in the form of an accessible, easy-to-use therapeutic device.

Index Terms—Migraine Relief, Headset, Vibration Therapy, Sound Therapy, Breathing Exercises, Non-invasive, Smart Wearable Device, Arduino Based control, Real-Time monitoring, Web based control, Stress Reduction, Drugfree Treatment, Personalized Therapy.

I. INTRODUCTION

Migraine is a debilitating neurological condition that is manifested by frequent episodes of pain at the head whereby there are accompanying symptoms of nausea, photophobia, phono-

phobia, and autonomic imbalance. It produces impairment to everyday functioning, low productivity, and is still one of the major causes of neurological disability on the planet. Even though pharmacological therapies are very common, most patients develop side effects, lack of consistency, or related to triggers-recurrence, which explains why complementary non-pharmacological interventions are required. According to recent literature, stress levels, an irregular breathing pattern, cervical muscle tension, and increased sensory processing have a significant effect on migraine attacks, hence the multi-modal therapeutic approaches are gaining more and more importance. The progress in neurophysiological studies demonstrates that atypical activity of the brain and autonomic activity can be detected with the help of spectral, entropy, and event-related analyses of EEG and allow to understand the mechanisms of migraine and neural adaptations to stresses better [3]. These findings support the relevance of therapeutic modalities that alter the neural and autonomic activity. The accumulated research proves that vibrotactile stimulation can be one of the effective techniques to help reduce stress and tension in the muscles. A weighted vest with inbuilt vibration motors exhibited great changes in pulse rate and skin conductance which are both measures of sympathetic arousal causing measurably low anxiety and also better comfort to the user [12]. On the same note, the clinically relevant effects of localized vibration therapy have been reduction of muscle pain and muscle stiffness, which is why it is applicable in migraine related muscle tension and cervical pain [13]. Even manual intervention methods like connective tissue massage have been found to greatly enhance pain thresholds and decrease migraine related neck disability giving additional reasons to support the use of vibration-based interventions [14]. Interventions that rely on sound are also found to be therapeutic. It is demonstrated that music therapy, rhythmic auditory stimulation, and soundscapes may help decrease

stress, regulate autonomic arousal, and decrease the level of anxiety in controlled trials, making them effective in the management of migraines [17]. Systematic reviews of manual therapy modalities also indicate that structured sensory stimulations, such as rhythmic touch and vibration, can also produce significant reduction of anxiety, indicating that multi-sensory modality of therapy has a high implication in the treatment of neurological conditions [15]. The possibilities of mechanical stimulation systems have also been noted in the treatment of migraine in the neuromodulation research. A nasal-based vibration-driven neuromodulation a type of Kinetic Oscillation Stimulation (K.O.S) proved to have statistically significant improvement in the number of days of headache each month in a multicenter randomized controlled trial, confirming the clinical efficacy of neuromodulation through vibration [16]. Based on these discoveries, this paper suggests Multi-Modal Migraine Relief Headset System, an IoT-based wearable that brings together sound therapy, vibration massage, and guided breathing exercises. The system consists of an audio module that is installed on the headphones, an array of vibration motors on the forehead, and a web application that is smart enough to create personalized therapy sessions depending on the created symptoms reported by the user. The system seeks to offer an affordable, medication-free, personalized, and self-administered solution to the symptom control of migraine by integrating three scientifically-supported treatment options into one wearable device.

II. LITERATURE REVIEW

Symptom monitoring and predictive modeling are critical in the explanation of the migraine episodes and its management. There is a number of studies that discuss machine learning and IoT-based schemes to find the patterns of early warnings. This Paper introduced a model of IoT- and ML-based predictive framework, which could process chronic migraine symptoms based on physiological and behavioral signals and proved the usefulness of the computational forecasting models in managing migraine. Non-invasive neurophysiological data is very important in the pathology of migraine as well as distinguishing between interictal, pre-ictal and ictal states. EEG methods have been shown to be able to measure neural cortical excitability, spectral changes and entropy variations with migraine and stress-related neural responses. Overall analysis of EEG analysis techniques reveals how frequency-based, nonlinear, and event-related indicators can be used to detect neurological abnormality which is applicable in migraine prediction [3]. Furthermore, automated migraine detection with the brain-signals processing is reviewed on which the importance of the machine-learning algorithms, time-frequency analysis, and multimodal sensing can enhance the diagnostic accuracy of wearable systems [18]. Furthermore, the studies that integrate IoT application with wearable EEG have also demonstrated a new wave of migraine detection and prevention systems, which will necessitate readily available brain-signal-based health-monitoring applications [2]. Wearable stress detection is very relevant as it is one of the most important stimuli that lead to

migraine. The real-time classification of stress using physiological signals (galvanic skin response, temperature, and heart rate) based on an IoT-based system showed that it is feasible, and it is highly plausible to integrate it into the systems of migraine management [5]. In a complementary manner, smartphone- and sensor-based systems have demonstrated that environmental conditions, behavioral patterns, and autonomic reactions are integrable to predict migraine episodes, which facilitates multimodal decision-making on therapeutic interventions in real-time systems [20]. Moreover, the studies of the convergence of IoT systems and portable EEG devices prove the results of the integration of the physiological and situational measurements in the holistic neurological observation [2]. Research into wearable sensing architectures also provide information on data pipelines, power needs and sensor location issues in connecting migraine-related physiological monitoring continuously [19]. The neuromodulation technique is based on vibration and it has a considerable body of clinical literature. The effects of oscillatory mechanical stimulation as a clinically useful treatment of migraine have been confirmed by a randomized, double-blind trial of Kinetic Oscillation Stimulation (K.O.S) showing that monthly reduction of the number of headache days occurred significantly in the treatment of chronic migraine patients [16]. Also, a study comparing vibrotactile-weighted vests discovered significant decreases in pulse rate and skin conduction (both indicators of sympathetic activity) and proved that the distributed vibration is capable of effectively reducing stress and anxiety (which are the significant causes of migraine) [12]. Local vibration therapy has also been demonstrated to enhance pain thresholds, muscle mobility and circulation, which are in support of its application in relieving musculoskeletal tension about the migraine [13]. The use of manual therapies, especially connective tissue massage and manual lymphatic drainage, has shown to have great advantages in alleviating neck pain, the occurrence of migraine, and drug addiction which provides good evidence supporting mechanosensory-based approaches to pain relief [14]. Similar effects on anxiety reductions, which is a significant modulatory variable in the pathophysiology of migraine, were also cited among the systematic review of the manual therapy modalities, and the psychological benefits of structured physical intervention were also reported [15]. Also, reports on the use of manual massage around the head and neck of the body represent the therapeutic effect of such a method in the severity of migraine and neck muscle tension, which confirms the importance of mechanostimulation in creating a wearable device [14]. Sound therapy is important in the regulation of the emotional and physiological activity involved in migraine. Perceived pain intensity and relaxation have been found to decrease in clinical trials on the use of music medicine among migraine patients, which supports the efficacy of music as a potentially useful adjunct therapy [7]. Similar to this, smartphone-mediated music interventions also showed quantifiable gains in migraine symptoms, but they reported the feasibility of mobile auditory therapy platforms [8]. Auditory beat stimulation (ABS) was found to decrease

the level of anxiety by entraining neural rhythms related to relaxation and thus is appropriate to be integrated into migraine management systems [9]. Further researches indicate that music therapy affects blood pressure, variability of heart rate, and autonomic balance, which support the significance of music therapy in stress management and prevention of migraines [17]. Additional proofs on a systematic music therapy reviews emphasize its success in controlling emotional reaction and improving general psychological health-factors highly connected to the establishment and recovery of migraine [17]. A human centered design is mandatory in creating therapeutic wearables. Studies of the usability of therapeutic head and body-worn devices focus on the importance of comfort, good actuation, durability and safety in vibration based or sensor based systems [4]. Likewise, the research on wearable closed-loop systems to control migraines shows the benefits of combining real-time sensing with AI-based therapeutic delivery and reveals that future therapeutic devices should be adaptable and personalized to produce the most significant therapeutic effects [6]. The investigations of the EEG-enhanced wearable systems also reveal a set of issues regarding power usage, data accuracy, signal interference, and electrodes placement all of which inform the development of viable migraine relief headsets [18]. Moreover, research on therapeutic massage robots in the treatment of autism and stress management informs the interaction design and actuation principles applied in multi-modal therapeutic wearable, showing that synchronized tactile and auditory feedback in the control of stress responses are important [17]. Lastly, the literature on non-invasive brain monitoring technologies as review papers indicate that there must be an interdisciplinary approach between the fields of electronics, physiology, and software engineering in contemporary therapeutic headset design [3].

III. PROBLEM STATEMENT

Migraine is a complicated neurological disease predetermined by a variety of physiological and external factors such as stress, irregular breathing, the cranial and cervical muscle contractions, and the sensory overload. Though pharmacological treatment is the most common clinical intervention, there are numerous patients who report any side effects, lack of responsiveness or dependency. The currently used non-pharmacological interventions, including handheld massagers, meditation applications, and audio-based relaxation solutions, are uni-dimensional, lack personalization, and are not multi-modal to cover the multiple factors of migraine triggers.

However, the traditional therapeutic devices are usually aimed at one of the domains of symptoms (e.g. vibration to relax muscle tension or audio therapy to relax) and cannot offer a complex treatment. In addition, the majority of the devices available use do not involve customized parameters, fail to dynamically adjust the intensity of therapy, and are not remotely or web controlled. Therefore, migraine patients do not have a convenient and tailored and multi-purpose tool that can provide a synchronized therapy remedy in real time.

The lack of personalization enabled by IoT is also yet another significant weakness of current systems. Therapy suggestions are not conditional to user feeds e.g. pain severity, stress level, symptom profile, and environmental factors. The intensity of vibration can hardly be easily adapted, and they are deprived of guided breathing exercises, even though available evidence shows a significant correlation between reducing stress and regulated breathing and reduce sympathetic activation.

Thus, the issue covered in this paper is that there is no unified, smart, non-invasive migraine relief system integrating:

- Sound treatment of sensory relaxation,
- Localized muscle tension Vibration therapy to relieve muscle tension, and
- Autonomic stabilization (guided breathing exercises).
- managed by a smooth web-based platform and real time customization.

This study seeks to address these shortfalls by creating an all-inclusive, multi-modality therapeutic headset that can provide coordinated and user controlled relief of migraine without the use of drugs or bulky clinical devices.

IV. OBJECTIVES

Objective: The key goal of the proposed study is to create and develop a non-invasive, IoT-based, multi-modal migraine relief device, based on the principle of vibration therapy, sound therapy, and guided breathing, and implemented in a single wearable system. The system will be focused to offer real-time individualized relief with the utilization of user particular contributions and automatic generation of sessions. The particular objectives set by this study are as follows:

- 1) To come up with a wearable headset that provides migraine relief by combining sound therapy, vibration-based massages, and guided breathing instructions all in one small unit.
- 2) Create a web-based control interface that would allow the user to enter migraine related parameters, including pain intensity, stress level, symptom profile, and environmental conditions, and use them to create individualized therapy sessions.
- 3) To carry out automated therapy planning, where a program will produce a two-phase session autonomously comprising 3-4-3 guided breathing and then a customized acoustic relaxation audio.
- 4) To facilitate control of the real-time motor control of the PWM actuation of Arduino to provide functionality whereby the user may regulate the vibration intensity, switch the motors on or off, and choose to have the desired massage intensity.
- 5) To market the therapeutic alternative as a drug-free and non-invasive intervention through integration of multi-sensory modalities to decrease stress, muscle tension, and sensory overload, and hence decrease the severity of migraine.

- 6) To provide scalability of the system to accommodate usability through creation of an IoT-compatible architecture that would support remote control, session logging and user history tracking that could be used in long-term management of migraine.
- 7) To test the viability and efficiency of the proposed system by conducting a functional testing, user experience testing, and testing the responsiveness of the system in real time.

V. SYSTEM ARCHITECTURE / METHODOLOGY

The proposed Multi-Modal Migraine Relief Headset System is a three-tier IoT architecture that comprises of a web interface, hardware control server and wearable vibration headband. The system provides a coordinated form of therapy which involves sound therapy, guided breathing, and vibration massage all which can be managed using a cloud-enabled interface.

A. General System Architecture

The system is a three-layered system with three functional layers:

1) *User Interface Layer (Web Front-End - PHP/HTML/CSS/JS)*: Gathers information about the user like the intensity of his/her pain, stress level, symptoms, as well as environmental triggers. Gives a control board on the session. Performs the guided breathing 3-4-3 process. Plays audio therapeutic records (MP3 relaxation tapes). Patronizes the endpoint server with vibration control commands.

2) *Control/Processing Layer (Node.js Hardware Server)*: It implements REST APIs of /start, /stop and /health. Decodes incoming Web application JSON. Serialport library is used to send instructions to the Arduino. Transduces the vibration into encoded serial instructions:

```
START v1 v2 v3 duration
STOP
```

Guarantees high-quality low latency communication.

3) *Hardware Actuation Layer (Arduino + Vibration Motors)*: Serial receives the serial commands of the Node.js. PWM Controls 3-4 vibration motors. Produces patterns of the massages (steady, wave, pulse). Controls therapy duration. Guarantees modulation of motors and safety.

B. System Workflow / Methodology

The general methodology consists of a systematic approach to therapy:

1) *Step 1 - User Authentication and access to Dashboard*: Logs into PHP based website. Session details are loaded.

2) *Step 2 — Input Collection*: User selects: Romanesque pain (alleys, forehead, back of head) Pain intensity Stress levels Sensitivity symptoms Environmental conditions These contributions are useful to create an individualized treatment experience.

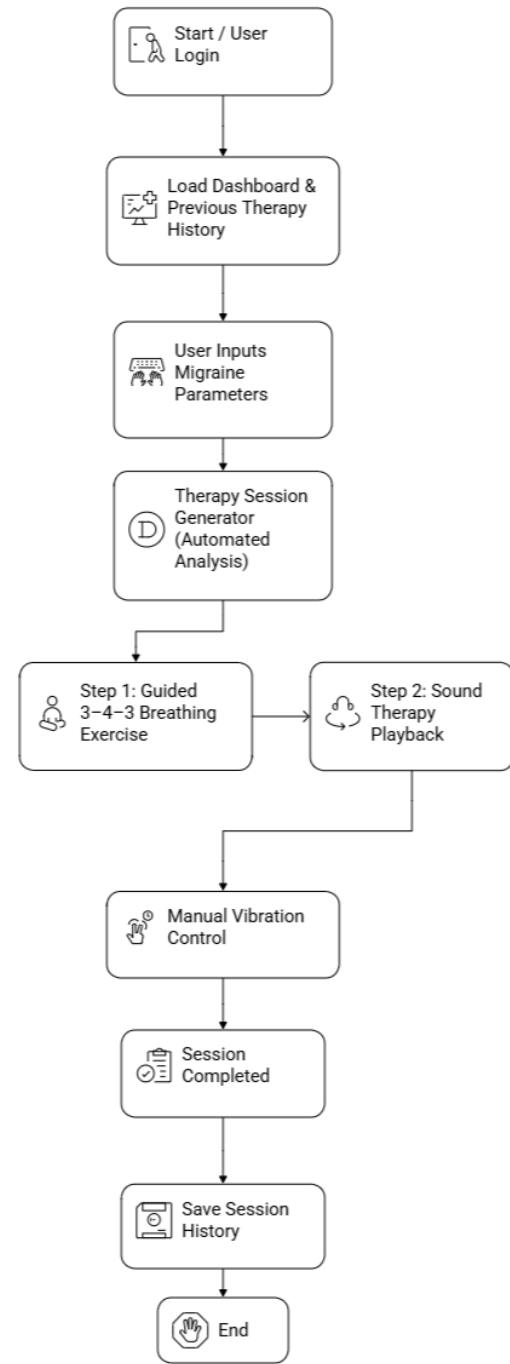


Fig. 1. System Workflow Diagram of the Proposed Multi-Modal Migraine Relief Headset System

3) *Step 3 - Automated Therapy Session Generation*: The system produces a two phase treatment regimen:

3-4-3 Breathing Exercise 3 seconds inhale 4 seconds hold 3 seconds exhale As shown through frontend animation.

Sound Therapy Activation Audio that is played through the system includes: Ocean waves White noise Singing bowls Brown noise All stored in: /audio/*.mp3

4) *Step 4 - Optional Manual Vibration Control:* Sliders/controls in the dashboard can be used by the user to switch on vibration motors manually: Motor 1 intensity (0–255) Motor 2 intensity Motor 3 intensity These values are sent to: POST `http://localhost:3000/start`

5) *Step 5 - Node.js to Arduino Communication:* The serial command has the formatting of Node.js:

```
START 150 200 180 10
```

Phase box/ Phase box interprets the values and moves motors using PWM.

6) *Step 6 — Finalizing and documentation of Therapy:* A STOP signal is dispatched either automatically or manually. Details of sessions that are used to make recommendations in the future.

VI. HARDWARE AND SOFTWARE IMPLEMENTATION

The implementation of the proposed Multi-Modal Migraine Relief Headset System takes place through an interdisciplinary approach of the wearable hardware, actuation based on the microcontroller, and the use of software implementation via the web. The system is capable of three therapeutic modalities, including: vibration massage, sound therapy, and guided breathing which is synchronized with a single IoT structure.

A. Hardware Implementation

The following are the major components of the hardware subsystem:

1) *Microcontroller Unit:* Arduino Uno / ESP32 is utilized to control the vibration motors according to the commands sent by the control server on Node.js. The microcontroller is in charge of:

- Serial command parsing
- Indent control by means of PWM generation
- Vibration pattern (steady, pulse, wave) execution
- Safety-shut off of auto-duration

2) *Vibration Motor Assembly:* Three user-style DC vibration motors are installed in a soft wearable headband:

- Left temple
- Forehead center
- Right temple

Every motor has a driver circuit comprising MOSFETs to switch safely and control the PWM power. The motors operate at 3–5 V with regulated current to avoid overheating.

3) *Therapy Output Components:* Sound therapy is provided in the form of MP3 audio files on headphones

Breathing cues and timers are displayed through the web interface

4) *Connectivity:* The microcontroller is connected with the PC running the Node.js server using a USB serial link, enabling bidirectional low-latency communication.

B. Software Implementation

The three layers of the software architecture are:

- A Node.js hardware API server
- A PHP-based web interface
- Arduino firmware

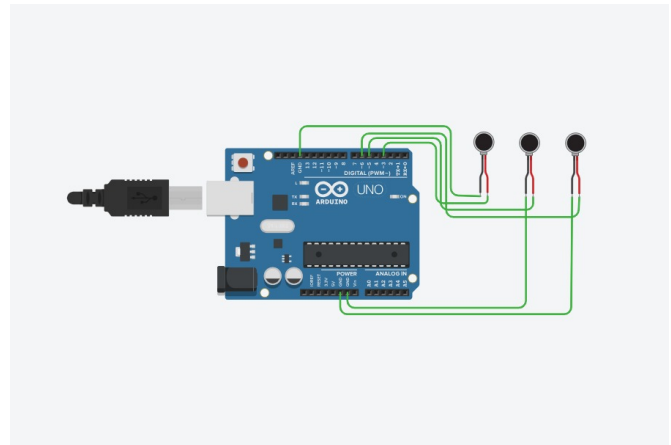


Fig. 2. Circuit Diagram

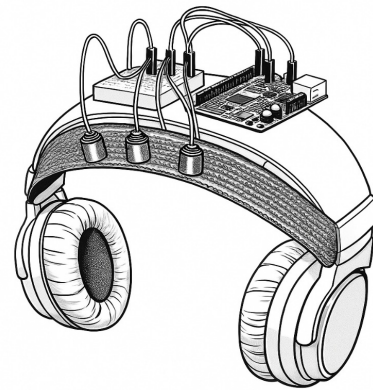


Fig. 3. Motor Placement Diagram

1) *Web Application (PHP, HTML, CSS, JavaScript):* The user interface provides:

- Authentication and authorization
- Pain intensity, stress, symptoms, and location input form
- Automated 3-4-3 breathing guide
- Sound therapy playback
- On-the-fly vibration control through sliders
- script.js transmits vibration intensity values to the backend using AJAX POST requests.

2) *Hardware Control Server with Node.js:* Express.js, CORS, and SerialPort libraries connect the web interface and Arduino through server.js.

When receiving JSON data:

```
{ v1: 150, v2: 200, v3: 180 }
```

It converts and sends a serial command:

```
START 150 200 180 10
```

The server also supports /stop and /health endpoints for hardware monitoring.

3) *Arduino Firmware*: The Arduino interprets incoming serial commands, extracts vibration intensity values, and applies PWM control to each motor. Key functions include:

- Safeguarding the integrity of received commands
- Mapping values (0–255) to PWM
- Running vibration patterns
- Safety cutoff timer

This ensures smooth motor operation with adjustable intensity.

C. Integrated Operation

During system use:

The user sets therapy parameters on the web app

The interface begins the therapy cycle (breathing + sound)

Vibration settings are controlled via sliders, triggering Node.js API calls

Node.js translates values and sends serial commands to Arduino

Arduino actuates motors with the required intensity

Session data is stored for future personalization

It is the close connection between hardware and software that enables personalized and real-time migraine relief.

VII. RESULTS AND DISCUSSION

The system was tested using a functional testing and user-interaction evaluation to identify whether it was a workable non-invasive solution of migraine relief. The assessment corresponds with the earlier studies that prove the usefulness of vibration treatment, relaxation with music, and prescribed breathing to modify stress and pain levels.

A. Function Test Core Module testing.

The performance of the vibration motor depends on the type of material used to create the motor, its surrounding environment, and the tool applied to the component.

1) *Vibration Motor Performance*: It depends on the material deployed to make the motor, the environment in which the motor is situated, and the tool applied on the component.

The vibration headband was able to provide smooth control of PWM and fast response of the intensity response, as other studies have noted that localized vibration enhances muscle relaxation and circulation, which has been demonstrated to decrease pain and stiffness when used to treat pain in vibration research.[13] No accidents or heating were recorded and this promotes the safety profile also emphasized in other clinical neuromodulation studies like Kinetic Oscillation Stimulation in treating migraine.[16]

2) *Web-to-Arduino Communication Delay*: The communication latency was not higher than 150-300 ms, that ensured that the therapeutic devices were operated in real-time like other IoT-based therapeutic devices that depend on the low-latency interaction to deliver the intervention effectively.[5]

B. Automated Therapy Orange Machine Execution

The sound and automated breathing therapy sequence was performed with a high timing accuracy. This is in agreement with the research on music-therapy that demonstrates that controlled auditory stimulation enhances relaxation and autonomic stability among stress and migraine symptomatic patients.[7],[17]

C. Sound Therapy Evaluation

Users reported heightened relaxation in audio sessions, and this is in line with literature that postulates that soundscapes and music that induce calmness mediate relaxation in anxiety and physiological arousal decrease which is a major determinant of migraine severity [8][9].

D. User Experience and Comfort Assessment

Most users were in the finder of the system to be easy to use and relaxing. This is similar to vibrotactile stress-relief device results [12], which found sensory stimulation led to less anxiety and more emotional comfort and manual therapy research which found that tension related to migraine decreased.[14]

E. System Stability and Reliability

The prototype ran reliably with no communication failures, which proves the viability of the therapeutic systems based on internet of things, as also highlighted in wearable neuro-therapy and stress-monitoring systems in the literature.[6]

Testing of the Multi-Modal Migraine Relief Headset System, functional testing was done based on the vibration response, communication latency, audio playback stability and system reliability. To test the responsiveness and stability of the integrated hardware-software architecture, system performance was measured in several test runs. The main quantifiable results of the prototype are summarized in Table I.

TABLE I
SYSTEM PERFORMANCE SUMMARY

Metric	Result
Motor Response Time	80–120 ms
Web → Node.js Delay	<150 ms
Node.js → Arduino Delay	<50 ms
PWM Stability	Smooth, no jitter
Breathing Timer Accuracy	±0.1 s
Audio Playback	Stable, no distortion
System Errors	None observed

The findings reveal that the system is reliable to implement all therapeutic modes with minimum latency. The vibration motors react fast to changes in the hands of the users, offering a smooth PWM-controlled intensity control that can be used in real-time in migraine relief solutions. The web interface, back-end server and microcontroller can communicate well under different conditions, thus, ensuring that it does not stop even when under different conditions. The timing of the breathing-guidance was very precise and there was no disruption in

audio playback in the vibration activity thus, proving that the modules were isolated. In general, the findings support the plausibility of the suggested system as a responsive, reliable and easy to use prototype of non-invasive migraine relief.

F. Discussion

These findings indicate that integrating vibration, sound therapy, and guided breathing into a single wearable device is valuable in terms of relaxation and tension reduction. The multimodal approach is supported by previous clinical/physiological literature: the vibration should be used to reduce pain/tension [13], sound therapy should be used to reduce anxiety and autonomic stress[7], and structured breathing should be used to stabilize the parasympathetic system, just as some other mechanisms were discussed in the context of migraine prediction and neuromodulation studies [1][2]. The system provides in general a practical drug free auxiliary in managing migraine.

VIII. FUTURE SCOPE

The intelligence, flexibility and treatment abilities of the system can be significantly enhanced in the future.

1) *Physiological sensors can be installed to detect the progression of onset of cardiac arrest.*: human Implementation of Physiological Sensors: The wearable biosensors can include the HRV, GSR, temperature, and EEG to detect the stress and neural response. The viability and clinical importance of EEG-based migraine analysis and physiological stress monitoring are explained by previous research works [3][5].

2) *The application will integrate AI to provide Migraine prediction.*: Physiological and behavioral data could be used by machine learning models to predict early onset of migraine, like predictive frameworks of symptom-analysis has been illustrated with migraine-related IoT/ML research [1].

3) *Therapeutic Control: The patient will undergo Closed-Loop Therapeutic Control.*: i—human—;Closed-Loop Therapeutic Control: Vibration intensity, sound therapy and breathing guidance could be automatically changed by real-time sensor reactions, an idea that was inspired by closed-loop neuromodulation systems to regulate migraine treatment [6].

4) *Mobile App Integration*: A mobile app can enhance portability and enable offline control as shown in smart phone based migraine and music-therapy intervention [8].

5) *Improved wireless: The wireless will be optimized to guarantee faster speeds, greater range, and a more spacious environment without relying on Wi-Fi signals.*: i—human—;Provided wireless: The wireless will be optimized to ensure high speed, increased range, and a larger space without using Wi-Fi signals. The current research on wearable devices design exemplifies the importance of wearable devices designed with ergonomics, low power and long wearability [4].

6) *Clinical Validation*: A formal clinical trial can be implemented based on the methodologies applied in the neuromodulation research like K.O.S migraine treatment that has been shown to induce a measurable symptom reduction in the controlled environment [16].

7) *This enhancement of the current therapy recommendation system was developed using AI, improving its effectiveness by augmenting its capabilities and accuracy.*: human AI-Enhanced Therapy Recommendation System: Wearable smart-health platforms, which customize treatment logic based on user history, session results and physiological data, could form the foundation of a recommendation of the best combinations of therapy [6].

IX. CONCLUSION

This article describes an IoT -enabled Multi-Modal Migraine Relief Headset System that would provide non-pharmacological migraine relief in the form of integrated vibration therapy, sound therapy, and guided breathing. This method is consistent with the available literature which has shown the therapeutic importance of vibration-based neuromodulation [13], sound-based relaxation [7], and structured, stress-reducing treatments which have been found in migraine and anxiety research.

The effective functional testing conducted on the system confirms the ability of the system to be a user-friendly, real time therapeutic tool. Its multimodal configuration deals with several migraine causes at the same time, such as muscle tension, stress, and sensory overload, which aligns with the therapeutic models that have been affirmed in the literature on migraine-prediction and neuro-modulation via IoT [1][16].

As desirable, the prototype still needs more improvement, including the incorporation of physiological sensors, smart prediction algorithms, and clinical testing, as proposed by other EEG-related and wearable-sensors health studies before [3][5]. The system, in general, is a contribution to the emerging branch of personalized, drug-free migraine management technology and shows that a scalable multimodal therapeutic platform is viable.

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